Maintenance Schedule

Gas Engine-Generator Set
mtu 8V4000 GS - 8V4000L64FNER
mtu 12V4000 GS - 12V4000L64FNER
mtu 16V4000 GS - 16V4000L64FNER
mtu 20V4000 GS - 20V4000L64FNER
Application group 3A

MS50298/03E

A Rolls-Royce solution

System designation		Engine model	Power per cylinder	Application group
Product name	Model number			
mtu 8V4000 GS	GG08V4000D1/D1M/D2	8V4000L64FNER (natural gas)	130 kW/cyl.	3A, continuous operation, unrestricted
mtu 12V4000 GS	GG12V4000D1/D1M/D2	12V4000L64FNER (natural gas)	130 kW/cyl.	3A, continuous operation, unrestricted
mtu 16V4000 GS	GG16V4000D1/D1M/D2	16V4000L64FNER (natural gas)	130 kW/cyl.	3A, continuous operation, unrestricted
mtu 20V4000 GS	GG20V4000D1/D1M/D2	20V4000L64FNER (natural gas)	130 kW/cyl.	3A, continuous operation, unrestricted
mtu 8V4000 GS	GG08V4000E1/E1M/E2	8V4000L64FNER (propane)	75 kW/cyl.	3A, continuous operation, unrestricted
mtu 12V4000 GS	GG12V4000E1/E1M/E2	12V4000L64FNER (propane)	75 kW/cyl.	3A, continuous operation, unrestricted
mtu 16V4000 GS	GG16V4000E1/E1M/E2	16V4000L64FNER (propane)	75 kW/cyl.	3A, continuous operation, unrestricted
mtu 20V4000 GS	GG20V4000E1/E1M/E2	20V4000L64FNER (propane)	75 kW/cyl.	3A, continuous operation, unrestricted

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for products from Rolls-Royce Solutions is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and are therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks must be carried out are specified in terms of operating hours and time limits. The limit which occurs first is applicable.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by a Rolls-Royce Solutions' Service Partner. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

This Maintenance Schedule and the intervals specified herein take effect as of initial commissioning.

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this Maintenance Schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carrying out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Checks by the operator

Important
Always observe the safety instructions.

Important
Non-observance of these instructions may lead to damage to the system!

Important
When the object of delivery is used abroad or in other special regions,Rolls-Royce Solutions is not responsible for observance of the statutory and other specifications / standards applicable at the place of use.

Recommended daily checks – System

- Check exhaust gas piping for condensate leaks (visual inspection)
- Check condensate collecting tank after condensate drain (if applicable)
- Carry out visual inspection of engine and plant for leaks / damage (bellows, hoses etc.)
- Check gas supply / gas system for damage (visual inspection)
- Checks on the SCR system (optional)

Only for 60 Hz transmission units

- Oil temperature
- Storage temperature
- Oil pressure
- Oil contamination indicator
- Leaks
- Oil level
- Noise, vibrations

Recommended daily checks – Control system

The following measured values and functions are monitored by the control system to protect the plant. These messages can also be called up via remote access (option) and can be monitored by the customer in the display.

- Electrical output
- Lube oil level
- Lube oil pressure
- Coolant water temperature
- Coolant water pressure
- Oxygen sensor voltage (option)
- Battery voltage
- Smooth running of engine
- Exhaust temperature
- Heating water supply and return temperature (option)
- Oil bleed system function (option)

Recommended weekly checks

- Lube oil consumption or content of fresh oil tank
- Engine room temperature
- Check of operating hours
- Check of number of starts
- Check of speed control for stability of regulator and the linkage (if installed)
- General visual inspection of module / engine-generator set
- Checks on the SCR system (optional)

Only for 60 Hz transmission units

TIM-ID: 0000097588 - 003

- Gearbox mounting
- Outer condition of transmission (contamination, oil deposits)
- Visual condition of the transmission oil

Additional checks

The operator's gas installation (upstream of twin solenoid valve) must be inspected for leaks at the following intervals:

- Every year for engines with turbocharger
- Every six months for aspirated engines

The following checks are to be carried out at appropriate intervals, depending on the degree of contamination:

- Check room air fan and fresh air filter for contamination
- Check filter or air-conditioning unit for control cabinet for contamination
- Check recooler for contamination

Electrical systems

Electrical systems and operating equipment must be checked prior to initial commissioning, after repairs, prior to re-commissioning and at regular intervals. The intervals must be arranged such that defects, which are to be expected, are detected at an early stage.

Here, among others, the following requirements apply:

- Accident prevention regulations BGV A3 "Electrical Systems and Operating Equipment"
- Technical rules for operational safety TRBS 2131 "Electrical Hazards"
- Industrial Safety Directive (BetrSichV)
- DIN VDE 0701/0702 Repairs, modifications and testing of electrical equipment/repeat tests on electrical units
- DIN EN 60204-1 Safety of machinery - Electrical equipment of machines

The operating company has the responsibility to see that these tests are carried out correctly. Electrical systems and fixed electrical operating equipment must be tested at appropriate intervals by a qualified electrician for their correct condition. The test is to include the following:

- Visual inspection and testing of protective equipment
- Testing of protective conductor resistance, insulation resistance, leakage current replacement, protective conductor current and differential current
- Carrying out fault diagnoses
- Preparation of test reports and inspection stickers

The industrial safety regulations, the technical rules for operational safety and the BGV A3 are fundamental with regard to adherence to the protective measures for "Electrical Systems and Operating Equipment". The operating company is accordingly obligated to care for the safety of electrical systems and operating equipment.

	Designation
DIN	Deutsches Institut für Normung (Federal German Standards Institute)
EN	Europäische Normung (European Standards)
ISO	International standard
VDE	Verband der Elektrotechnik, Elektronik und Informationstechnik (Association for electrical engineering, electronics and information technology)
BGV	Berufsgenossenschaftliche Vorschriften (Trade association regulations)
TRBS	Technische Regeln für Betriebssicherheit (Technical rules for operational safety)

Table 2: Designations for standards and regulations

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Important

These regulations are contained in the manufacturers' instructions, statutory regulations and technical guidelines valid in the individual countries. Since there can be major differences from country to country, it is not possible to provide a universally applicable statement on the rules to be observed within the framework of this maintenance schedule.
The user of the product mentioned herein is obligated to inform him-/herself about the provisions applicable in his/her country. Rolls-Royce Solutions accepts no liability whatsoever for improper or illegal use of the electrical system, which it has approved.

TIM-ID: 0000097588 - 003

1.4 Additional notes on the maintenance of gas systems

General information

The maintenance schedule matrix normally finishes with extended component maintenance or with engine overhaul. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the task list serve as a reference for the scope of parts required.

Important
Repairs tasks on the gas system, for which the gas-conducting area is opened, and leak or stress tests must only be carried out by specialized companies. As a rule these are companies that are approved for the gas industry with qualified personnel who have been trained for technical gas applications.

Important
Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in Fluids and Lubricants Specifications and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet Rolls-Royce Solutions specifications and be approved by the relevant component manufacturer.

Among other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Gas filter:
  The maintenance interval depends on gaseous fuel quality (purity). Fuel filters upstream of the engine must be regularly cleaned and renewed if necessary.
- Battery:
  Battery maintenance depends on duty and ambient conditions. The specifications of the battery manufacturer apply.

Items listed in this maintenance schedule but which are not included in the scope of supply may be disregarded.

Generator maintenance

Refer to the Operating Instructions provided by the manufacturer for the intervals for these maintenance tasks.

Generator circuit breaker maintenance

Refer to the Operating Instructions provided by the manufacturer for the intervals for these maintenance tasks.

Maintenance of non-MTU components

Refer to the Operating Instructions provided by the relevant manufacturer for the intervals for these maintenance tasks.

Maintenance tasks

The respective maintenance tasks to be carried out are based on operating hours (see the following pages).

Number of starts

In general, each engine start is to be counted for the number of engine starts.

TIM-ID: 0000085431 - 005

1.5 Allocation matrix for application group 3A

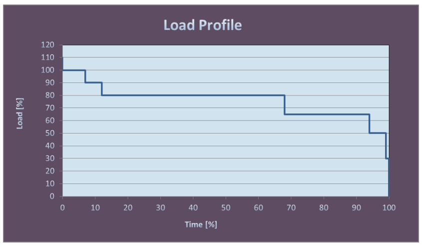
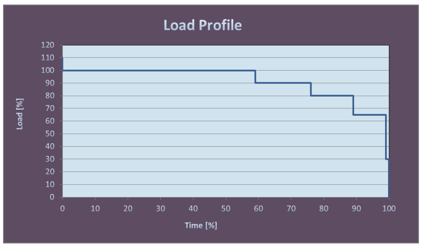
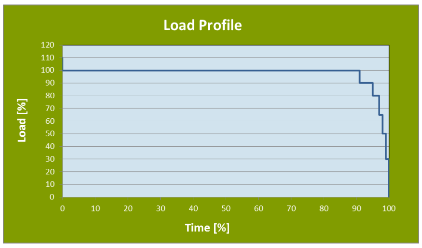
Load profile: Load %	Time %	Load factor %	Load indicator %	130 kW/cyl. natural gas	75 kW/cyl. propane	|	|	
110 (100-110)	—	≤ 85	≤ 60	84,000 28,000 (→ 1.7)	—	—	—	
100 (90-100)	7							
90 (80-90)	5							
80 (65-80)	56							
65 (50-65)	26							
50 (30-50)	5							
30 (5-30)	1							
Idle speed (0-5)	—							
110 (100-110)	—	≤ 95	≤ 85	84,000 24,500 (→ 1.8)	—	—	—	
100 (90-100)	59							
90 (80-90)	17							
80 (65-80)	13							
65 (50-65)	10							
50 (30-50)	—							
30 (5-30)	1							
Idle speed (0-5)	—							
110 (100-110)	—	≤100	≤100	84,000 21,000 (→ 1.9)	84,000 28,000 (→ 1.7)	—	—	
100 (90-100)	91							
90 (80-90)	4							
80 (65-80)	2							
65 (50-65)	1							
50 (30-50)	1							
30 (5-30)	1							
Idle speed (0-5)	—							

Table 3: Allocation matrix for application group 3A

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